

Husnain Nadeem

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PROFESSIONAL SUMMARY

Computer Science undergraduate (National Textile University, CGPA 3.74/4.00) with research interests in cloud networking, distributed systems, and AI-driven data analysis. Independently designed and executed a multi-region cloud network performance study on AWS EC2, combining automated measurement, Python-based data analysis, and applied AI experimentation. Motivated to contribute to research-driven projects in cloud computing and networked systems.

AREA OF EXPERTISE

Cloud Networking · Distributed Systems · Network Performance Analysis · Data Analysis · Python Automation · Artificial Intelligence Applications · Cloud Computing

EDUCATION

Bachelor of Science in Computer Science (BSCS)

Oct 2023 – July 2027

National Textile University, Pakistan CGPA: 3.74/4.00

PROFESSIONAL TRAINING

HCCDP – Solution Architectures Cloud Developer Professional

(In Progress, 2026)

Center for Skill Faisalabad (C4S) — 6-Month Professional Training Program

- Training in enterprise cloud architecture, Linux administration, routing and switching, and the interconnection between cloud services and applications.
- Applying concepts through independent AWS and DevOps projects alongside coursework.

RESEARCH EXPERIENCE

Cloud Network Performance Benchmarking Across AWS Regions

2026

Independent Research AWS EC2, Python, ping, iperf3

- Designed a multi-region AWS EC2 study to measure cloud network performance across geographically distributed regions.
- Built a Python-based automated pipeline (ping, iperf3) to collect latency, jitter, packet loss, throughput, and TCP retransmission data.
- Structured measurements into CSV datasets and performed statistical analysis and visualization in Python.
- Investigated variation in network behavior across cloud regions, establishing a foundation for further experiments under varying load conditions.

Command-to-Video: AI-Assisted Generative Media Pipeline

2025

Independent AI Research Project Python, Hugging Face Diffusers, Gemini API

- Built a pipeline converting natural-language commands into short, generated videos, using an LLM API (Gemini) for prompt refinement and a Hugging Face diffusion model (Stable Diffusion) for text-to-image generation.
- Evaluated model and API choices for prompt-to-image quality, then automated frame sequencing into video output.
- Explored practical trade-offs in AI-generated image/video workflows, including inference speed, image consistency, and API-based prompt engineering.

PROJECTS

- **Cloud-Native DevOps Portfolio:** Deployed a CloudFront/S3-hosted portfolio with an automated GitHub Actions CI/CD pipeline and cache invalidation.
- **End-to-End CI/CD Pipeline (Jenkins, Argo CD, Kubernetes):** Built a GitOps pipeline to AWS EKS; automated build, test, and deployment stages.
- **Multi-AZ AWS Production Architecture:** Designed a fault-tolerant VPC with auto scaling, load balancing, and bastion-controlled access.
- **AWS-Native CI/CD Pipeline Implemented:** Code Pipeline/Code Build/Code Deploy workflow with IAM-based service permissions.
- **Infrastructure as Code (CloudFormation): Provisioned** and audited AWS resources via reusable YAML templates with drift detection.
- **Cloud Cost Optimization Automation:** Serverless (Lambda/CloudWatch) workflow to detect and clean up unused AWS resources.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

Cloud & Infrastructure AWS (EC2, S3, VPC, IAM, RDS, Lambda, Serverless, CloudWatch, App Mesh) · Terraform · Docker & Docker Compose · CI/CD concepts

Networking: TCP/IP, Routing, Switching, Network Performance Analysis

Data & Research: Pandas, Matplotlib, Data Analysis

Web: HTML, CSS, JavaScript

LEADERSHIP & ACTIVITIES

- External Affairs, Computer Science Society, National Textile University (2025–Present) manage outreach and partnership activities.
- DCS Ambassador, Quran Society, National Textile University (2026–Present) contribute to student and community engagement.

CERTIFICATIONS

- AWS Cloud Practitioner Essentials Amazon Web Services, 2025
- Job Roles in the Cloud Amazon Web Services, 2025